# Lecture 2: PAC learning

## Machine Learning 2026

Federico Chiariotti (federico.chiariotti@unipd.it)



Università degli Studi di Padova

# Recap

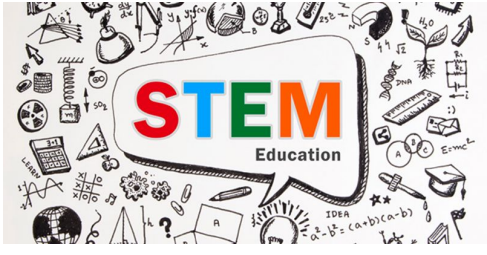


Università
degli Studi
di Padova

# Elearning

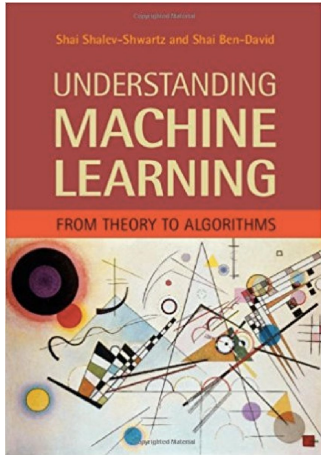
ALL course material and announcements will be published through the course Elearning page. The password for manual enrollment is **ML2627PW** – but most of you should be enrolled automatically.

https://stem.elearning.unipd.it

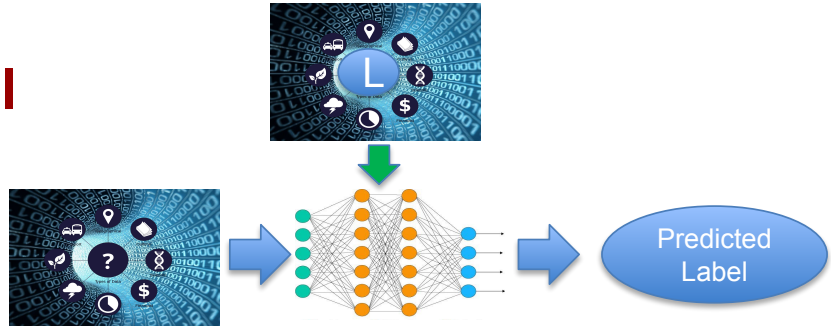


# Course material

- Main book: Shalev-Shwartz, Shai, & Ben-David, Shai, *Understanding machine learning: From theory to algorithms*, Cambridge University Press, 2014
- PDF available from the authors at http://www.cs.huji.ac.il/~shais/UnderstandingMachineLearning/copy.html
- Lecture slides: available on Elearning
- Lab solutions and notebooks: available on Elearning



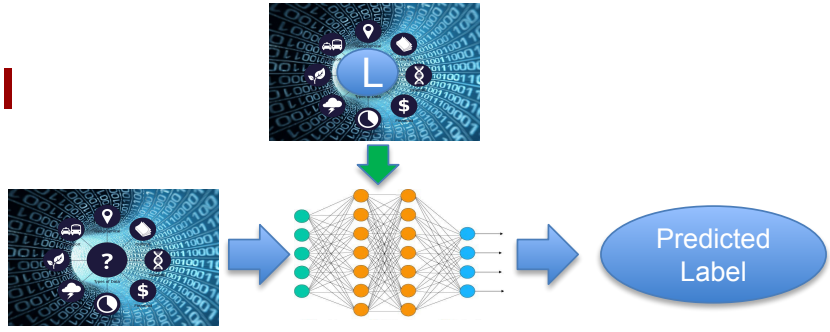
# Supervised learning model




- Domain set (or instance space) $X$
  - $x \in X$ is a domain point or instance
  - Usually, $x$ is represented by a tensor of *features*
- Label set $Y$
  - Simplest case: binary classification, $Y = \{0, 1\}$
- Training set $S$
  - Finite sequence of labeled points $((x_1, y_1), (x_2, y_2), \ldots, (x_m, y_m))$
  - Each point $x_i$ is associated to its label $y_i$

Assumpt: the labels are correct

# Supervised learning model




- Prediction rule $h : X \to Y$
  - The learner's output (hypothesis)
  - $A(S)$: hypothesis produced by algorithm $A$ when it is fed training set $S$
- Data generation model
  - $D$ is a distribution over $X$ **(unknown to the machine learning algorithm)**
  - Instances are labeled according to $f : X \to Y$ **(unknown to the ML algorithm)**
  - Training set: sampling according to $D, y_i = f(x_i)\ \forall x_i \in S$
- Success metric
  - Classifier error: probability of predicting the wrong label over $D$

from which we draw the points

Also, in real world, D (≡ frequency of a data) is unknown

We have to find it!

Sometimes for testing you test an f on things you know the answer

He talked about assumption I didn't hear (~10 min)

error = $\begin{cases} 0 \text{ for } r \text{ ensm.} \\ 1 \text{ for } x \text{ "} \end{cases}$ ← Per Y binary, there are other ones

You select it based on what you need

# Loss functions

- ➔ Assume a domain subset $A \subset X$
- ➔ $A$ is an event expressed by $\pi : X \to \{0, 1\}$, i.e., $A = \{x \in X : \pi(x) = 1\}$
- ➔ We get $P_{x \sim D}[\pi(x) = 1] = D(A)$

Error of prediction rule $h : X \to Y$

$$L_{D,f}(h) \stackrel{\text{def}}{=} P_{x \sim D}[h(x) \neq f(x)] = D(x : h(x) \neq f(x))$$

Loss function: probability that, given $x_i$, $f(x_i) \neq y_i$ => Prediction ≠ label

$L_{D,f}(h)$ (often written just as $L_D(h)$) is called true loss, true risk, or generalization loss

But you don't know f and D

# Empirical Risk Minimization (ERM)

- ➔ Learner outputs $h_S : X \to Y$
- ➔ **Goal: find $\mathbf{h}^*$ that minimizes $\mathbf{L_{D,f}(h)}$**
- ➔ Both $D$ and $f$ are unknown!

ERM: we minimize the loss on the training set $\overline{L_S(h)} = \frac{\sum_{i=1}^{m} |h(x_i) - y_i|}{m}$

# of mistakes, normalized over D: 0 = everything right, 1 = everything wrong

This works if we use a 0-1 loss: the definition can be generalized

The empirical risk is also called training error or training loss

Problem: sometimes the D is not "general distribution", "being unlucky"

# Hypothesis classes

We can apply ERM over a *restricted* class $H$ of possible hypotheses

$h \in H$ is a function $h : X \to Y$

Restricted ERM finds $h^* \in \arg\min_{h \in H} L_S(h)$

There might be multiple solutions with the same empirical risk

How can we choose a class that avoids overfitting?

# Assumptions

1. Let us start with a *finite* class $H$
2. Let $h_S = \mathrm{ERM}_H \in \arg\min_{h \in H} L_S(h)$

Idea: $h_S$ is the one with lower empyrical risk (works on $L_S(h)$)
$h^*$ è la perfetta, minimizza $L_{D,f}$ ($L_{D,f}(h^*) = 0$)
↑ l'assunzione è che $h^*$ esiste e sia nelle nostre hyp...
=> Aim = $h^* = h_S$

We need two further assumptions:

We picked the right class of hypothesis => $h^* \in$ set of hypothesis => At least one function is perfect

3. We are lucky, and have *realizability*: $f \in H$, i.e., $\exists h^* \in H : L_{D,f}(h^*) = 0$
4. IID assumption: samples are drawn independently from $D$

Can we learn $h^*$ with ERM?

# Part 1:
# PAC learning



Università
degli Studi
di Padova

# PAC learning

the more data the smaller $\varepsilon$

PAC stands for **probably approximately correct**

1. **Accuracy**: we are happy with a loss $L_{D,f}(h_S) < \varepsilon$ $L_{D,f}(h_S) \neq 0$ it cannot $\Rightarrow$ we approximate $< \varepsilon$
2. **Confidence**: we can never fully rule out unlucky training sets, but we can limit their probability to a given $\delta$

Assuming iid for the training set

$0 < \varepsilon, \delta < 1$ (sono probabilità)

$|H|$ := cardinality => how many inside # of hypothesis

# PAC learning theorem

La epsilon ti garantisce una approssimazione per hs molto buona, vicina a h star. Delta ti garantisce che la probabilità che quello avvenga è molto alta!

Let $H$ be a **finite** hypothesis class. Let $\delta, \varepsilon \in (0,1)$ and $m \in \mathbb{N}$ such that:

$$m \geq \frac{1}{\varepsilon} \log\left(\frac{|H|}{\delta}\right)$$

- Inversely proportional to $\varepsilon$
- Logarithmic growth with the size of $H$ and $1/\delta$

if that's true => our $h_S$ is PAC safe => Approximately correct with a high probability for

For any $D$ and $f$ for which **realizability holds**, we have that any $h_S$ computed with ERM over training set $S$ of size $m$ respects $L_{D,f}(h_S) \leq \varepsilon$ with probability greater than $1 - \delta$

$$M = \{S \in X^m : \exists h \in H_B, L_S(h) = 0\}$$

$$M = \bigcup_{h \in H_B} \{S \in X^m : L_S(h) = 0\}$$

**Dimostrazione** ← ognuna sarà così e nelle slide

$H_B \subset H$

$H_B = \{ h \in H : L_{D,f}(h) > \varepsilon \}$ ← This is a bad set of hypothesis

$M = \{ S \in X^m : \exists h \in H_B, L_S(h) = 0 \}$ ($X^m$ = spazi di dim m; training)

$M$ = insieme di training set che con una brutta h comunque danno $L_S(h)=0$ ⇒ Caso in cui h molto buona su S ⇒ sappiamo andare male su X ($L_{D,f}(h)>\varepsilon$)

"M sono tutti gli S con"

⇒ set of misleading training set

$P(L_{D,f}(h_S) > \varepsilon) \leq D^m(M)$ ← probability of having a bad training set = prob di pescare da M

O: if we pick a bad t.s. then $L_S(h^*)=0$ ($h^*$ = good h)

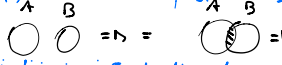
We don't know which one ERM will pick!

Proof matematica: 1 accade ⇔ 2 accade → Idea: devo prima avere un S e h, solo allora posso avere $L_S(h)=0 \wedge L_{D,f}(h)>\varepsilon$ (h bad)

**Note**

prob(M) $p(\bigcup_{h \in H_B} \dots)$ $\Rightarrow$ prob(M) ≤ prob(2) $p(L_S(h)=0)$

$D(A \cup B) \leq D(A) + D(B)$ perché




⇒ = ⇒ < perché l'intersez si calcola 2 volte

Ora: Solo riscritture "trovate le $h_S$, unisci gli insiemi S che li contengono" → ossia: cambi h → trovi $M_h$ → trovi altro $M_h$ → … → $\bigcup_{h \in H_B}$

$M = \bigcup_{h \in H_B} \{ S \in X^m : L_S(h)=0 \}$ ($M_h$) ⇒ $D^m(M) \leq \sum_{h \in H_B} D^m(M_h)$ ← Stessa idea:




$\bigcup_{h \in H_B} M_h$ ← le ≤ sono contate più volte

$L_S(h)=0 \Leftrightarrow h(x_i) = f(x_i)\ \forall i$ ⇒ $D^m(\{S : L_S(h)=0\}) = \prod_{i=1}^{m} D(\{x_i : h(x_i) = f(x_i)\}) < (1-\varepsilon)^m$ (∀ dim)

lo stesso S ha più pt $x_i$ t.c. …

$\equiv 1 - L_{D,f}(h)$ $L_{D,f}(h) > \varepsilon$ def

per def $L_{D,f}(h) = P(h(x_i) \neq f(x_i))$ 2nd $1 - L_{D,f}(h) < 1 - \varepsilon$

$\Rightarrow |\bigcup_h M_h| < |M|$

$\Rightarrow \sum_{h \in H_B} D^m(M_h) < D^m(M)$

⇒ Rivedi su prima parte: $D^m(M) \leq \sum_{h \in H_B} D^m(M_h) \leq \sum_{h \in H_B} (1-\varepsilon)^m \leq e^{-\varepsilon m}$ ← $(1-x \leq e^{-x})$

per → Tells why it's strict

$L_{D,f}(h) < \delta$ da prima e da def

Now: $P(L_{D,f}(h_S) > \varepsilon) \leq D^m(M) \leq |H_B| e^{-\varepsilon m} < |H| e^{-\varepsilon m}$ and $\delta \geq |H| e^{-\varepsilon m} \Rightarrow \frac{\delta}{|H|} \geq e^{-\varepsilon m} \Rightarrow \log\left(\frac{\delta}{|H|}\right) \geq \varepsilon m$

$\Rightarrow -m \leq -\frac{1}{\varepsilon} \log\left(\frac{|H|}{\delta}\right)$

$m \geq \frac{1}{\varepsilon} \log\left(\frac{|H|}{\delta}\right)$

c.v.d.

We have a log curve ⇒ H ↗ ⇒ m →

* le ≤ : $h_i \neq h_j$ ma $h_i$ e $h_j$ possono arrivare dagli stessi S

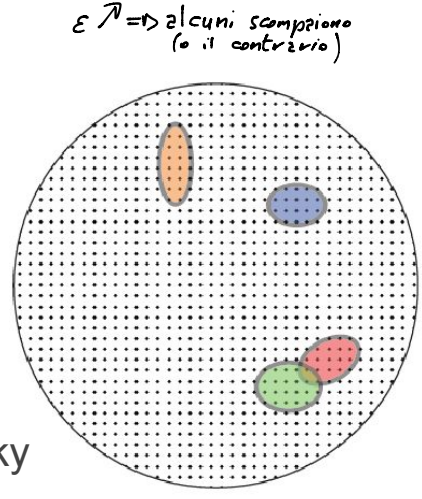
# Graphical idea

Let us define the set of *bad hypotheses* $H_B = \{h \in H : L_{D,f}(h) > \varepsilon\}$

We can consider each possible training set $S$ as a point in the circle. The colored areas are sets that lead to $h_S \in H_B$

- ➔ Increasing $m$ leads to a wider circle (fewer unlucky training sets
- ➔ Increasing $\varepsilon$ narrows down $H_B$

The theorem places a bound on the probability of being unlucky



# Proof

Let us define the set of training sets in which a bad hypothesis works

$$M = \{S \in X^m : \exists h \in H_B, L_S(h) = 0\}$$

We know that, in order for ERM to fail, we must be in $M$ (remember, ERM selects one of the hypotheses with zero training loss):

$$p(L_{D,f}(h_S) > \varepsilon) \leq D^m(M)$$

IID hypothesis

# The union bound

Let us consider two event sets $A$ and $B$ and distribution $D$. We have

$$D(A \cup B) \leq D(A) + D(B)$$

This is easy to prove: the union operation includes the intersection only once, so if an outcome is in both $A$ and $B$ it gets counted twice on the right side

# First step

We can rewrite set $M$ as

$$M = \bigcup_{h \in H_B} \{S \in X^m : L_S(h) = 0\}$$

We can apply the union bound:

$$p(L_{D,f}(h_S) > \varepsilon) \leq D^m(M) \leq \sum_{h \in H_B} D^m(\{S : L_S(h) = 0\})$$

# Second step

Saying that a hypothesis has zero training loss is equivalent to saying it correctly classifies all points in the training set:

$$D^m(\{S : L_S(h) = 0\}) = \prod_{i=1}^{m} D(\{x_i : h(x_i) = f(x_i)\})$$

By the definition of a bad hypothesis, we have

$$D(\{x : h(x) = f(x)\}) < 1 - \varepsilon$$

As the sample is IID, we know that

$$D^m(\{S : L_S(h) = 0\}) < (1 - \varepsilon)^m$$

# Third step

As the sample is IID, we know that

$$D^m(\{S : L_S(h) = 0\}) < (1 - \varepsilon)^m$$

We can use inequality $(1 - x) \leq e^{-x}$ to get

$$D^m(\{S : L_S(h) = 0\}) < e^{-\varepsilon m}$$

Since we need to sum over $H_B$, we get

$$D^m(M) < |H_B|e^{-\varepsilon m} < |H|e^{-\varepsilon m}$$

Realizability: at least one good hypothesis

# Putting it all together

We just proved that

$$p(L_{D,f}(h_S) > \varepsilon) \leq D^m(M) < |H|e^{-\varepsilon m}$$

We can then guarantee condition $p(L_{D,f}(h_S) > \varepsilon) < \delta$ if

$$\delta > |H|e^{-\varepsilon m}$$

This is equivalent to the theorem statement

$$m \geq \frac{1}{\varepsilon} \log \left( \frac{|H|}{\delta} \right)$$

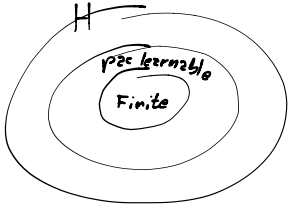
# PAC learnability

A hypothesis class $H$ is PAC learnable if there is a function $m_H : (0,1)^2 \to \mathbb{N}$ and a learning algorithm such that for every $\delta, \varepsilon \in (0,1)$ and for every learning problem $D, f$ with the realizability assumption, the algorithm will satisfy the PAC condition over an IID training set of size $m \geq m_H$ ❶

So, if I pick $\varepsilon, \delta$, I have ❶,... $\Rightarrow$ PAC learnable

$m \geq \frac{1}{\varepsilon} \log\left(\frac{|H|}{\delta}\right)$ is sufficient to be PAC learable

So: $|H| < \infty \not\Rightarrow$ PAC learnable + being PAC tells us that $\exists m_H$



# Corollary

Any finite $H$ is PAC learnable, as we just proved the existence of a function that satisfies the requirement:

Picked $\varepsilon, \delta$ always $\exists m$

$$m \geq \frac{1}{\varepsilon} \log \left( \frac{|H|}{\delta} \right)$$

However, this is a sufficient, not a necessary condition: some infinite classes might still be PAC learnable!

# Remarks

- ➔ PAC learnability is a property of the hypothesis class
- ➔ It must hold for all functions and distributions
- ➔ It requires the realizability assumption (strong!)

Aim, changin $|H|$ with complexity of $h$ (=> from linear to quadratic)

If we follow the finite hypothesis bound, we see that larger classes require larger training sets, as does restricting the conditions to be more accurate or more confident

# Part 2:
# Generalizing our results

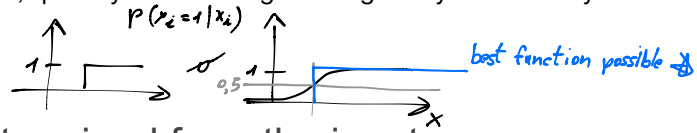
Idea: we have very limiting assumptions



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

# Adding noise

Example: students in high school with higher grades in math will go to a scientific degree in uni. But not everyone, quindi your x is not guaratiing always the same y. There is noise then!




In real life, labels are not always perfectly determined from the inputs

⇒ D becomes a distribution over $X \times Y$

We can replace $f$ by considering a joint distribution $D$ over $X \times Y$

The problem is to learn the marginal distribution $D(y|x)$

In this case, we do not have realizability (if the label is random, we cannot get 0 loss)

# Bayesian prediction

If we know $D$, we can directly identify the optimal predictor by applying Bayes' rule:

$$f_D(x) = \begin{cases} 0, & \text{if } P(y=1|x) < 0.5; \\ 1, & \text{otherwise.} \end{cases}$$

This predictor is not achievable in practice (it would require knowing the solution in advance), but it represents an upper bound to learning performance

# Agnostic PAC learnability

Since we dropped the realizability assumption, we cannot reach 0 loss. What we can try and guarantee is a bound on the loss with respect to the minimum possible loss in the hypothesis class

A hypothesis class $H$ is **agnostic** PAC learnable if there is a function $m_H : (0,1)^2 \to \mathbb{N}$ and a learning algorithm such that for every $\delta, \varepsilon \in (0,1)$ and for every learning problem $D$, the algorithm will satisfy the following condition over an IID training set of size $m \geq m_H$:

w/o this would be PAC

"minimum true loss over H"

$$L_D(h) \leq \left[\min_{h^* \in H} L_D(h^*)\right] + \varepsilon$$

"We can get arbitrely close to the best approximation"
riascoltalo!

# A more general learning problem


Image by Steffen Prößdorf, CC BY-SA 4.0

1. Binary classification: $Y = \{0, 1\}$

   Is it a goalkeeper or a field player?

More real case scenario

2. Multiclass classification: $Y = \{0, 1, \ldots, K-1\}$ ($K$ = # of classes)

   Is it a keeper, defender, midfielder, or striker?

Still same thing, but with K outputs, not binary

3. Regression: $Y \subseteq \mathbb{R}$

   The target set is a subset of real numbers (given her physical features, what is her value on the market?)

It will be very different for any of this problem

How can we adapt our loss function?

# Generalizing the loss function

- ➔ Hypothesis class $H$
- ➔ Domain $Z = X \times Y$

Any function $\ell : H \times Z \to \mathbb{R}^+$ can be a valid loss function

Risk function: expected loss of a hypothesis $h \in H$

$$L_D(h) = \mathbb{E}_{(x,y)\sim D}[\ell(h,(x,y))]$$

Empirical risk: average loss over the training set $S$

$$L_S(h) = \frac{1}{m}\sum_{i=1}^{m} \ell(h,(x_i,y_i))$$

# Common loss functions

- 0-1 loss: commonly used in classification

$$\ell_{0-1}(h,(x,y)) \triangleq I(h(x) \neq y)$$

- Cross-entropy: often used in deep learning classification (we will study it later in the course)
- L2 (squared) loss:

$$\ell_{\text{sq}}(h,(x,y)) \triangleq (h(x)-y)^2$$

- L1 (linear) loss:

$$\ell_{\text{lin}}(h,(x,y)) \triangleq |h(x)-y|$$

Loss functions are application-dependent, but computational cost also plays a part

# Agnostic PAC learnability (generalized loss)

A hypothesis class $H$ is **agnostic** PAC learnable if there is a function $m_H : (0,1)^2 \to \mathbb{N}$ and a learning algorithm such that for every $\delta, \varepsilon \in (0,1)$ and for every learning problem $D$, the algorithm will satisfy the following condition over an IID training set of size $m \geq m_H$:

$$L_D(h) \leq \min_{h^* \in H} L_D(h^*) + \varepsilon$$

$$L_D(h) = \mathbb{E}_{(x,y)\sim D}[\ell(h,(x,y))]$$

It' not part of problem, it's connected to what you want to work

# Finding the right loss function

The same application (e.g., fingerprint-based identification) can have different loss functions depending on the context:

- ➔ If the fingerprint is used to let subscribers into a gym, the worst that can happen is that a person trains without paying (and customers at a high-end gym do not want the hassle of scanning multiple times or getting locked out)
- ➔ If the fingerprint is used to access dangerous chemicals, the consequences for a false positive are very serious

Idea: depending on the situation, it changes how much and what you care about
and that's the creative part